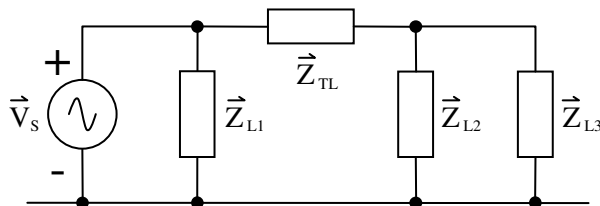


Problem #1

A power system equivalent circuit and the values of its impedance parameters are given below. **Calculate the equivalent impedance of the circuit seen by the source.** Show your final answer below in polar form with a precision of **5 decimal places in the magnitude** and **3 decimal places in the phase angle**.



$$\begin{aligned}\bar{Z}_{L1} &= 100 \angle 30^\circ \Omega & \bar{Z}_{TL} &= 0.987 \angle 80^\circ \Omega \\ \bar{Z}_{L2} &= 300 \angle -60^\circ \Omega \\ \bar{Z}_{L3} &= 900 \angle 70^\circ \Omega\end{aligned}$$

$$\bar{Z}_{eqS} = \left\{ \bar{Z}_{L1}^{-1} + \left[\bar{Z}_{TL} + \left(\bar{Z}_{L2}^{-1} + \bar{Z}_{L3}^{-1} \right)^{-1} \right]^{-1} \right\}^{-1} = 89.52336 \angle 16.430^\circ \Omega$$

Problem #2

Repeat the previous problem using the new value for the equivalent impedance of Load #2 given below.

$$\bar{Z}_{L2} = 300 \angle 20^\circ \Omega$$

$$\bar{Z}_{eqS} = \left\{ \bar{Z}_{L1}^{-1} + \left[\bar{Z}_{TL} + \left(\bar{Z}_{L2}^{-1} + \bar{Z}_{L3}^{-1} \right)^{-1} \right]^{-1} \right\}^{-1} = 70.80678 \angle 30.598^\circ \Omega$$

Problem #3

Perform the following calculations. Show only your final answer below in polar form with a precision of **6 decimal places in the magnitude** and **3 decimal places in the phase angle**.

$$R = 0.4321, \quad GMR = 0.0375$$

$$GMD_{eq} = \sqrt[3]{(16)(12)\sqrt{16^2 + 12^2}}$$

$$L = 3.218688 \times 10^{-4} \cdot \ln(GMD_{eq}/GMR)$$

$$\vec{Z} = R + j120\pi L = 0.850223 \angle 59.455^\circ$$

Problem #4

A single phase 965V source supplies 160KVA with a 0.83 lagging power factor to a constant impedance load. Compute the **real and reactive power supplied by the source** and the **source current**.

$$1\phi, \quad V_s = 965 \text{ V}, \quad S_s = 160 \text{ KVA}, \quad pf_s = 0.83 \text{ lagging} \rightarrow \theta_s = +\cos^{-1} 0.83$$

$$P_s = S_s \cos \theta_s = S_s \cdot pf_s = 132.8000 \text{ KW}$$

$$S_s = V_s I_s \rightarrow I_s = S_s / V_s = 165.803 \text{ A}$$

$$Q_s = S_s \sin \theta_s = 89.2421 \text{ KVAR}$$

Problem #5

A three phase 4160V Δ -connected source supplies 60A with a 0.75 leading power factor to a Y-connected constant impedance load. Compute the **real and reactive power supplied by the source**.

$$3\phi, \quad V_{SLL} = 4160 \text{ V}, \quad I_{SL} = 60 \text{ A}, \quad pf_s = 0.75 \text{ leading} \rightarrow \theta_s = -\cos^{-1} 0.75$$

$$P_{S3\phi} = \sqrt{3} \cdot V_{SLL} I_{SL} \cos \theta_s = \sqrt{3} \cdot V_{SLL} I_{SL} \cdot pf_s = 324.2399 \text{ KW}$$

$$Q_{S3\phi} = \sqrt{3} \cdot V_{SLL} I_{SL} \sin \theta_s = -285.9527 \text{ KVAR}$$